CO2 Emissions Analysis - Business Case Study

1. Data Import and Initial Exploration

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import scipy.stats as stats
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score, mean_absolute_error
from sklearn.feature_selection import SelectKBest, f_regression
```

```
# Load the data set
df = pd.read_csv('CO2_Emissions.csv')
# df.info()
```

```
df.shape
```

```
(7385, 12)
```

```
df.head()
```

	Make	Model	Vehicle Class	Engine Size(L)	Cylinders	Transmission	Fuel Type	Fuel Consumption City (L/100 km)	Fuel Consumption Hwy (L/100 km)	Fuel Consumption Comb (L/100 km)	Consumpt Comb (r
0	ACURA	ILX	COMPACT	2.0	4	AS5	Z	9.9	6.7	8.5	
1	ACURA	ILX	COMPACT	2.4	4	M6	Z	11.2	7.7	9.6	
2	ACURA	ILX HYBRID	COMPACT	1.5	4	AV7	Z	6.0	5.8	5.9	
3	ACURA	MDX	SUV -	3.5	6	AS6	Z	12.7	9.1	11.1	

Next steps: Generate code with df New interactive sheet

```
df.tail()
```

	Make	Model	Vehicle Class	Engine Size(L)	Cylinders	Transmission	Fuel Type	Fuel Consumption City (L/100 km)	Fuel Consumption Hwy (L/100 km)	Fuel Consumption Comb (L/100 km)	Consur Comb
7380	VOLVO	XC40 T5 AWD	SUV - SMALL	2.0	4	AS8	Z	10.7	7.7	9.4	
7381	VOLVO	XC60 T5 AWD	SUV - SMALL	2.0	4	AS8	Z	11.2	8.3	9.9	
7382	VOLVO	XC60 T6 AWD	SUV - SMALL	2.0	4	AS8	Z	11.7	8.6	10.3	

2. Data Quality Aassessment and Cleaning

Examine the dataset for data quality issues including missing values, duplicates.

```
# Check if there are any missing data
missing_data = df.isnull().sum()
# missing_data
```

```
# Check any duplicate data
duplicate_data = df.duplicated().sum()
duplicate_data
```

```
np.int64(1103)
```

```
# Remove the duplicates
df_cleaned = df.drop_duplicates()
df_cleaned.shape
```

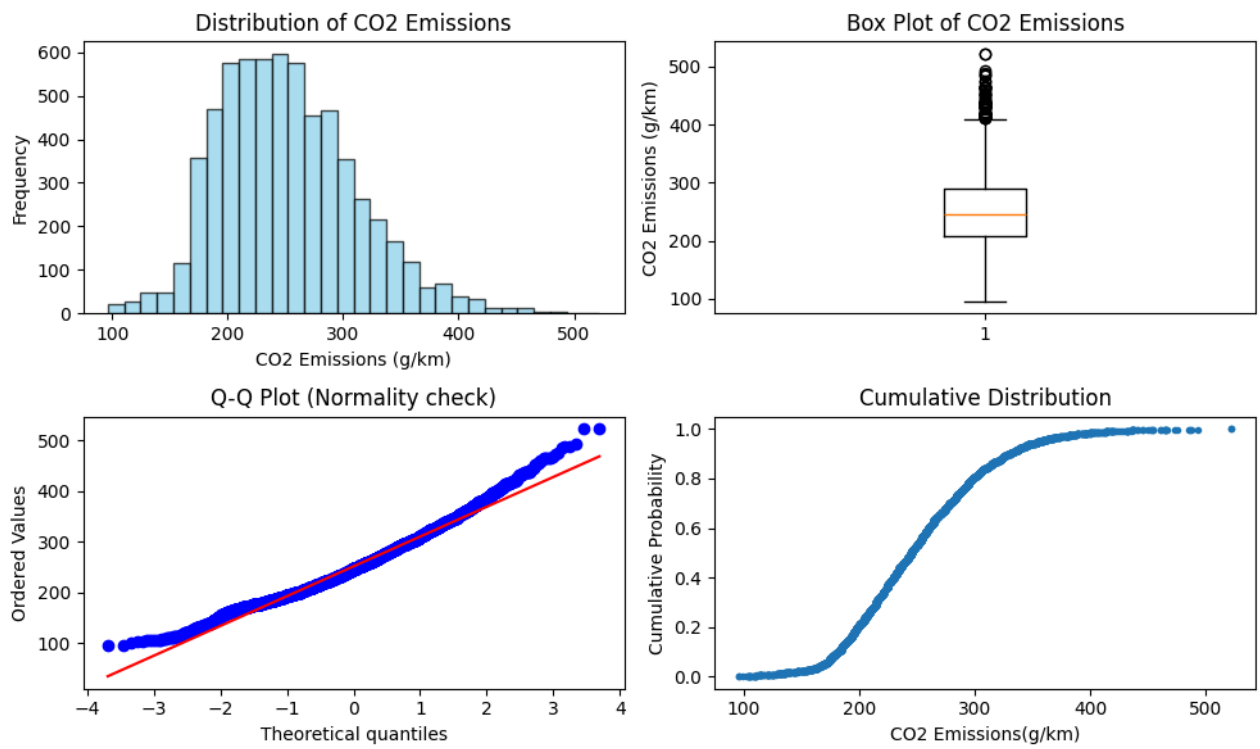
```
(7385, 12)
```

```
df_final = df_cleaned.copy()
# Target column
target_column = 'CO2 Emissions(g/km)'
```

3. Exploratory Data Analysis (EDA)

Understand the distribution of variables and identify key characteristics of our CO2 emissions data.

```
fig, axes = plt.subplots(nrows=2, ncols=2, figsize=(10, 6))

# Histogram
axes[0,0].hist(df_final[target_column], bins=30, alpha=0.7, color='skyblue',
               edgecolor='black')
axes[0,0].set_title('Distribution of CO2 Emissions')
axes[0,0].set_xlabel('CO2 Emissions (g/km)')
axes[0,0].set_ylabel('Frequency')

# Box plot
axes[0,1].boxplot(df_final[target_column])
axes[0,1].set_title('Box Plot of CO2 Emissions')
axes[0,1].set_ylabel('CO2 Emissions (g/km)')

# Quantile-Quantile plot for normality
stats.probplot(df_final[target_column], dist="norm", plot=axes[1,0])
axes[1,0].set_title('Q-Q Plot (Normality check)')

# Cumulative distribution
sorted_values = np.sort(df_final[target_column])
y = np.arange(1, len(sorted_values)+1) / len(sorted_values)
axes[1,1].plot(sorted_values, y, marker='.', linestyle='none')
axes[1,1].set_title('Cumulative Distribution')
axes[1,1].set_xlabel(target_column)
axes[1,1].set_ylabel('Cumulative Probability')

plt.tight_layout()
plt.show()
```



```
# Get the Mean, Median, SD and Skewness
mean = df_final[target_column].mean()
median = df_final[target_column].median()
sd = df_final[target_column].std()
skewness = df_final[target_column].skew()

print(f"Mean: {mean:.2f}")
print(f"Median: {median:.2f}")
print(f"Standard Deviation: {sd:.2f}")
print(f"Skewness: {skewness:.2f}")
```

```
Mean: 251.16
Median: 246.00
Standard Deviation: 59.29
Skewness: 0.56
```

```
# Get categorical variables
categorical_columns = df_final.select_dtypes(include=['object']).columns
categorical_columns
```

```
Index(['Make', 'Model', 'Vehicle Class', 'Transmission', 'Fuel Type'], dtype='object')
```

```
# Get numerical variables
numerical_columns = df_final.select_dtypes(include=['int64', 'float64']).columns
numerical_columns
```

```
Index(['Engine Size(L)', 'Cylinders', 'Fuel Consumption City (L/100 km)',
       'Fuel Consumption Hwy (L/100 km)', 'Fuel Consumption Comb (L/100 km)',
       'Fuel Consumption Comb (mpg)', 'CO2 Emissions(g/km)'],
      dtype='object')
```

4. Correlations

Relationships between variables and identify which features have the strongest correlation with CO2 emissions.

```
# Correlations metrics
correlation_matrix = df_final[numerical_columns].corr()
correlation_matrix
```

	Engine Size(L)	Cylinders	Fuel Consumption City (L/100 km)	Fuel Consumption Hwy (L/100 km)	Fuel Consumption Comb (L/100 km)	Fuel Consumption Comb (mpg)	CO2 Emissions(g/km)
Engine Size(L)	1.000000	0.928843	0.834228	0.766817	0.820146	-0.762955	0.854802
Cylinders	0.928843	1.000000	0.801277	0.717647	0.781099	-0.723731	0.834687
Fuel Consumption City (L/100 km)	0.834228	0.801277	1.000000	0.950811	0.994052	-0.927640	0.918756
Fuel Consumption Hwy (L/100 km)	0.766817	0.717647	0.950811	1.000000	0.978607	-0.891892	0.883424

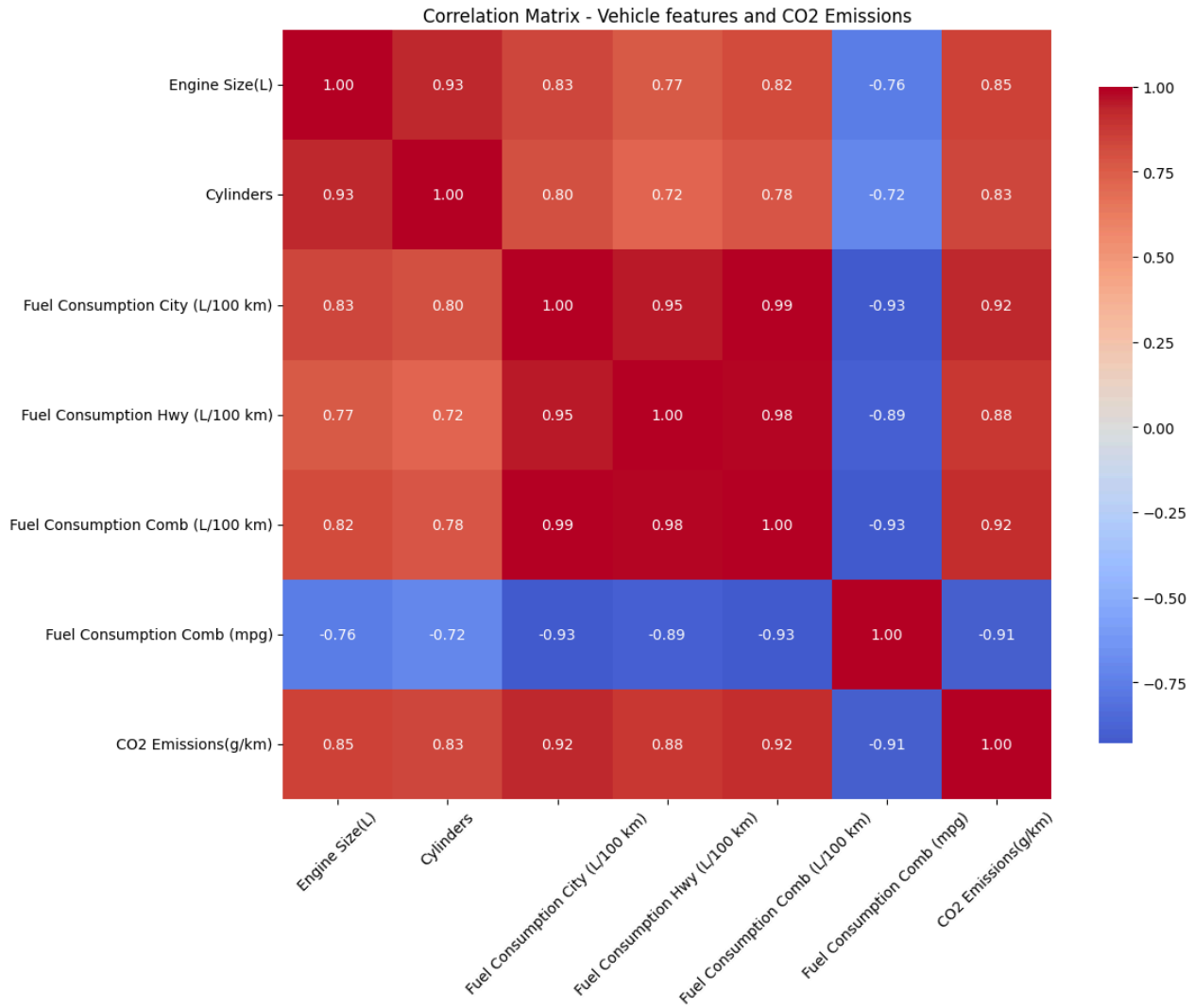
Next steps: Generate code with `correlation_matrix` New interactive sheet

```
# Correlations with target variable
target_correlations = correlation_matrix[target_column].sort_values(
    ascending=False, key=abs)
target_correlations
```

	CO2 Emissions(g/km)
CO2 Emissions(g/km)	1.000000
Fuel Consumption City (L/100 km)	0.918756
Fuel Consumption Comb (L/100 km)	0.916840
Fuel Consumption Comb (mpg)	-0.906783
Fuel Consumption Hwy (L/100 km)	0.883424
Engine Size(L)	0.854802
Cylinders	0.834687

dtype: float64

```
# Correlation heatmap
plt.figure(figsize=(12, 10))
# mask = np.triu(np.ones_like(correlation_matrix, dtype=bool))
sns.heatmap(correlation_matrix,
            annot=True,
            # mask=mask,
            cmap='coolwarm',
            fmt='.2f',
            center=0,
            square=True,
            cbar_kws={'shrink':0.8})
plt.title('Correlation Matrix - Vehicle features and CO2 Emissions')

plt.xticks(rotation=45)
plt.yticks(rotation=0)

plt.tight_layout()
plt.show()
```




```
# Identify TOP Predictors .. top 5
top_predictors = target_correlations.drop(target_column).head(5)
top_predictors
```

	CO2 Emissions(g/km)
Fuel Consumption City (L/100 km)	0.918756
Fuel Consumption Comb (L/100 km)	0.916840
Fuel Consumption Comb (mpg)	-0.906783
Fuel Consumption Hwy (L/100 km)	0.883424
Engine Size(L)	0.854802

dtype: float64

```
# Statistical Significance tests for correlations

from scipy.stats import pearsonr

significant_correlations = []
for col in numerical_columns:
  if col != target_column:
    corr_coef, p_value = pearsonr(df_final[col], df_final[target_column])
    significance = '***' if p_value < 0.001 else '**' if p_value < 0.01 else '*' if p_value < 0.05 else 'NS'
    significant_correlations.append({'Variable':col,
        'Correlation': corr_coef,
        'P-Value': p_value,
```

```
        'Significance': significance
    })
sig_df = pd.DataFrame(significant_correlations).sort_values('Correlation',
                                                            ascending=False,
                                                            key=abs)
display(sig_df)
```

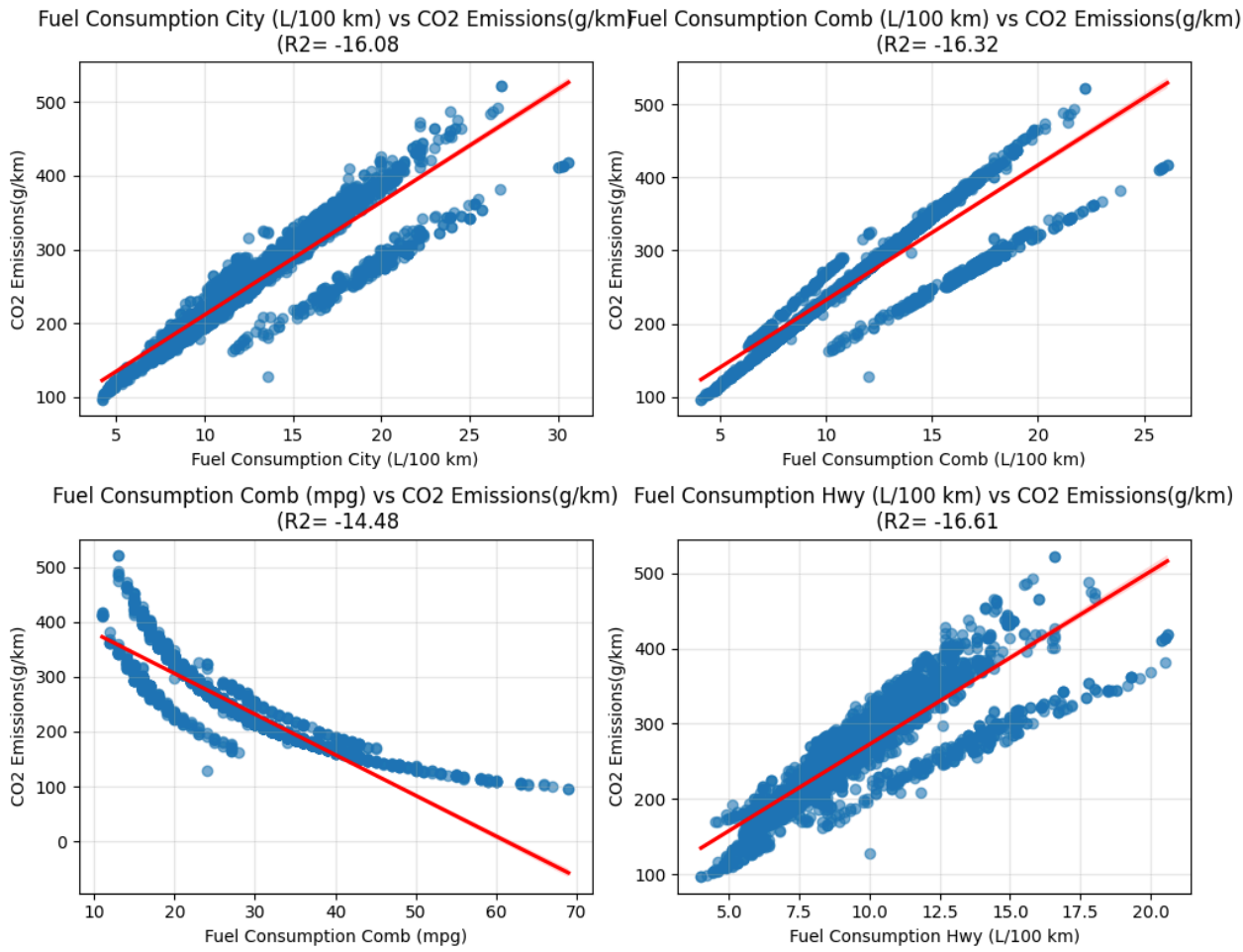
	Variable	Correlation	P-Value	Significance
2	Fuel Consumption City (L/100 km)	0.918756	0.0	***
4	Fuel Consumption Comb (L/100 km)	0.916840	0.0	***
5	Fuel Consumption Comb (mpg)	-0.906783	0.0	***
3	Fuel Consumption Hwy (L/100 km)	0.883424	0.0	***
0	Engine Size(L)	0.854802	0.0	***
1	Cylinders	0.834687	0.0	***

Next steps: Generate code with `sig_df` New interactive sheet

5. Data Visulaization and Pattern Discovery

Create visualizations to uncover patterns and relationships between vehicle features and CO2 emissions.

```
# Scatter plots for top numerical predictors
top_numerical_predictors = [col for col in top_predictors.index[:4] if col in numerical_columns]

fig, axes = plt.subplots(2, 2, figsize=(10, 8))
axes = axes.ravel()

for i, predictor in enumerate(top_numerical_predictors):
  if i < 4:
    # scatter plot with regression line
    sns.regplot(x=predictor,
                y=target_column,
                data=df_final,
                ax=axes[i],
                scatter_kws={'alpha':0.6},
                line_kws={'color':'red'})
    # Calculate R2 (R2 square)
    r2 = r2_score(df_final[target_column], df_final[predictor])
    axes[i].set_title(f'{predictor} vs {target_column}\n (R2= {r2:.2f}')
    axes[i].grid(True, alpha=0.3)

plt.tight_layout()
plt.show()
```




```
Start coding or generate with AI.
```

6. Categorical Analysis and Comparision

Analyze how CO2 emissions vary across different categorical variables such as vehicle class, fuel type, and transmission type.

```
# Analyse CO2 Emissions by Categorical Variables
categorical_analysis = {}

for categorical_column in categorical_columns:
  category_stats = df_final.groupby(categorical_column)[target_column].agg(
      ['mean', 'median', 'std', 'count', 'min', 'max']).round(2)
  categorical_analysis[categorical_column] = category_stats

  # I disaplyed only top 5
  display(category_stats.sort_values(by='mean', ascending=False).head(5))
```

Make	mean	median	std	count	min	max
BUGATTI	522.00	522.0	0.00	2	522	522
LAMBORGHINI	402.27	384.0	50.32	37	331	493
SRT	389.00	389.0	0.00	2	389	389
ROLLS-ROYCE	388.75	393.0	12.32	48	359	404
BENTLEY	367.91	373.0	41.16	35	291	465

Model	mean	median	std	count	min	max
Chiron	522.0	522.0	NaN	1	522	522
CHIRON	522.0	522.0	NaN	1	522	522
Aventador Roadster	493.0	493.0	NaN	1	493	493
E350 WAGON	488.0	488.0	NaN	1	488	488
Aventador Coupe	486.0	486.0	1.41	2	485	487

Vehicle Class	mean	median	std	count	min	max
VAN - PASSENGER	397.21	414.0	42.27	66	315	488
VAN - CARGO	361.50	362.0	17.87	22	334	396
SUV - STANDARD	306.73	306.0	44.30	613	156	476
PICKUP TRUCK - STANDARD	301.61	298.0	31.04	475	238	414
TWO-SEATER	282.09	270.0	75.11	381	145	522

Transmission	mean	median	std	count	min	max
A7	314.27	291.0	53.26	44	248	473
A10	313.96	325.5	32.13	28	238	357
A5	299.37	299.0	42.63	78	172	488
A4	296.59	344.0	83.91	61	168	421
A8	285.42	286.0	43.11	378	181	413

Fuel Type	mean	median	std	count	min	max
E	276.05	280.0	48.74	330	128	418
Z	265.73	254.0	57.64	2765	122	522
X	235.98	229.0	58.30	3039	96	488
D	235.24	243.0	41.91	147	169	326
N	213.00	213.0	NaN	1	213	213

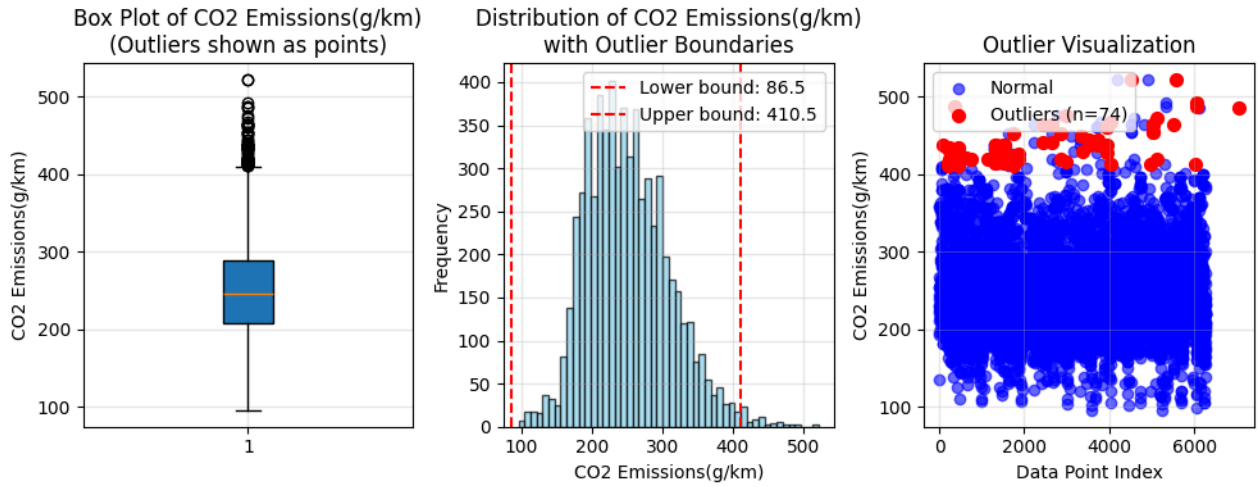
7. Outlier Detection and Analysis

Identify vehicles with unusually high or low CO2 emissions and investigate their characteristics.

```
# Outlier detection using IQR method
def detect_outliers_iqr(data, column):
    Q1 = data[column].quantile(0.25)
    Q3 = data[column].quantile(0.75)
    IQR = Q3 - Q1
    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR

    outliers = data[(data[column] < lower_bound) | (data[column] > upper_bound)]
    return outliers, lower_bound, upper_bound

# Outlier detection using Z-score method
def detect_outliers_zscore(data, column, threshold=3):
    z_scores = np.abs(stats.zscore(data[column]))
    outliers = data[z_scores > threshold]
    return outliers
```

```
# IQR method
iqr_outliers, lower_bound, upper_bound = detect_outliers_iqr(df_final,
                                                              target_column)

# Z-score method
zscore_outliers = detect_outliers_zscore(df_final, target_column)
```

```
# Visualize outliers
fig, axes = plt.subplots(1, 3, figsize=(10, 4))

# Box plot highlighting outliers
axes[0].boxplot(df_final[target_column], patch_artist=True)
axes[0].set_title(f'Box Plot of {target_column}\n(Outliers shown as points)')
axes[0].set_ylabel(target_column)
axes[0].grid(True, alpha=0.3)

# Histogram with outlier boundaries
axes[1].hist(df_final[target_column],
             bins=50, alpha=0.7, color='skyblue', edgecolor='black')
axes[1].axvline(lower_bound, color='red',
                linestyle='--', label=f'Lower bound: {lower_bound:.1f}')
axes[1].axvline(upper_bound, color='red',
                linestyle='--', label=f'Upper bound: {upper_bound:.1f}')
axes[1].set_title(f'Distribution of {target_column}\nwith Outlier Boundaries')
axes[1].set_xlabel(target_column)
axes[1].set_ylabel('Frequency')
axes[1].legend()
axes[1].grid(True, alpha=0.3)

# Scatter plot showing outliers
axes[2].scatter(range(len(df_final)),
                df_final[target_column],
                alpha=0.6, color='blue', label='Normal')
outlier_indices = iqr_outliers.index
axes[2].scatter(outlier_indices, df_final.loc[outlier_indices, target_column],
               color='red', s=50, label=f'Outliers (n={len(outlier_indices)})')
axes[2].set_title('Outlier Visualization')
axes[2].set_xlabel('Data Point Index')
axes[2].set_ylabel(target_column)
axes[2].legend()
axes[2].grid(True, alpha=0.3)

plt.tight_layout()
plt.show()
```



Start coding or generate with AI.

8. Data Preprocessing for Modelling

Prepare the dataset for machine learning by encoding categorical variables.

```
df_model = df_final.copy()

# Identify features and target
```

```
feature_cols = [col for col in df_model.columns if col != target_column]
X = df_model[feature_cols].copy()
y = df_model[target_column].copy()
```

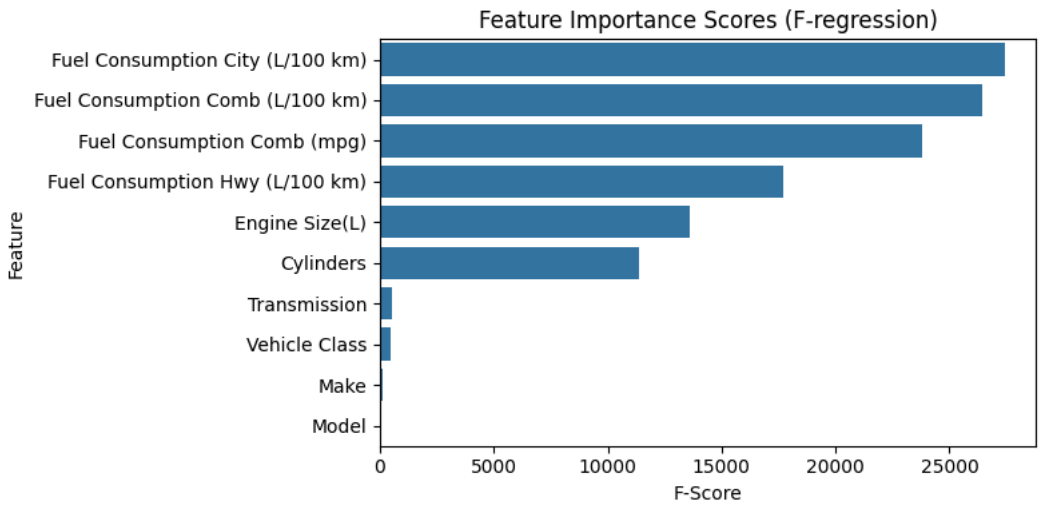
```
# Handle categorical variables
categorical_encoders = {}
X_processed = X.copy()

for cat_col in categorical_columns:
    if cat_col in X_processed.columns:

        # Use Label Encoding for simplicity (could use One-Hot Encoding for better results)
        le = LabelEncoder()
        X_processed[cat_col] = le.fit_transform(X_processed[cat_col])
        categorical_encoders[cat_col] = le
```

9. Model Development and Training

Develop a linear regression model to predict CO2 emissions and understand feature importance.

```
# Split data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
  X_processed, y, test_size=0.2, random_state=42, stratify=None
)

# Use SelectKBest with f_regression to identify top features
n_features_to_select = min(10, X_train.shape[1])  # Select top 10 or all available features
selector = SelectKBest(score_func=f_regression, k=n_features_to_select)
X_train_selected = selector.fit_transform(X_train, y_train)
X_test_selected = selector.transform(X_test)

# Get selected feature names
selected_features = X_train.columns[selector.get_support()].tolist()
feature_scores = selector.scores_[selector.get_support()]

feature_importance_df = pd.DataFrame({
    'Feature': selected_features,
    'Importance Score': feature_scores
}).sort_values('Importance Score', ascending=False)

display(feature_importance_df)

# Visualize feature importance
plt.figure(figsize=(8, 4))
sns.barplot(data=feature_importance_df, x='Importance Score', y='Feature')
plt.title('Feature Importance Scores (F-regression)')
plt.xlabel('F-Score')
plt.tight_layout()
plt.show()
```

	Feature	Importance Score
6	Fuel Consumption City (L/100 km)	27416.057907
8	Fuel Consumption Comb (L/100 km)	26470.683119
9	Fuel Consumption Comb (mpg)	23799.333624
7	Fuel Consumption Hwy (L/100 km)	17690.206793
3	Engine Size(L)	13606.616725
4	Cylinders	11401.571569
5	Transmission	551.049266
2	Vehicle Class	449.369815
0	Make	109.934053
1	Model	59.759131



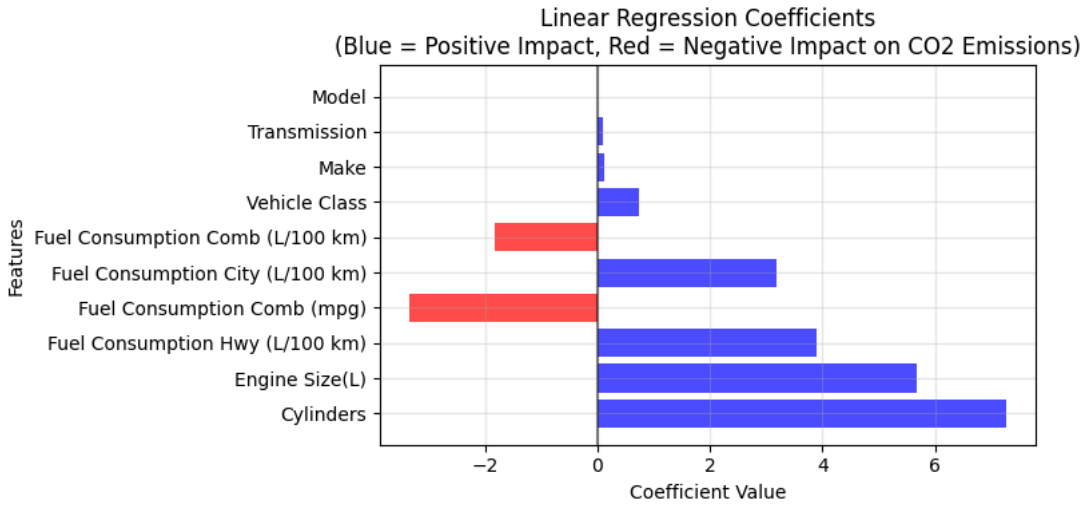
Next steps: Generate code with feature_importance_df New interactive sheet

```
# TRAIN LINEAR REGRESSION MODEL

# Train model with all features
lr_full = LinearRegression()
lr_full.fit(X_train, y_train)

# Train model with selected features
lr_selected = LinearRegression()
lr_selected.fit(X_train_selected, y_train)

coefficients_df = pd.DataFrame({
    'Feature': selected_features,
    'Coefficient': lr_selected.coef_
}).sort_values('Coefficient', ascending=False, key=abs)

display(coefficients_df)
```

	Feature	Coefficient
4	Cylinders	7.262583
3	Engine Size(L)	5.674658
7	Fuel Consumption Hwy (L/100 km)	3.885396
9	Fuel Consumption Comb (mpg)	-3.344644
6	Fuel Consumption City (L/100 km)	3.194615
8	Fuel Consumption Comb (L/100 km)	-1.839836
2	Vehicle Class	0.739459
0	Make	0.111751
5	Transmission	0.095042
1	Model	-0.000127

Next steps: Generate code with coefficients_df New interactive sheet

```
# Visualize model coefficients
plt.figure(figsize=(8, 4))

# Create a horizontal bar plot
colors = ['red' if coef < 0 else 'blue' for coef in coefficients_df['Coefficient']]
plt.barh(coefficients_df['Feature'], coefficients_df['Coefficient'], color=colors, alpha=0.7)

plt.title('Linear Regression Coefficients\n(Blue = Positive Impact, Red = Negative Impact on CO2 Emissions)')
plt.xlabel('Coefficient Value')
plt.ylabel('Features')
plt.axvline(x=0, color='black', linestyle='-', alpha=0.5)
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```



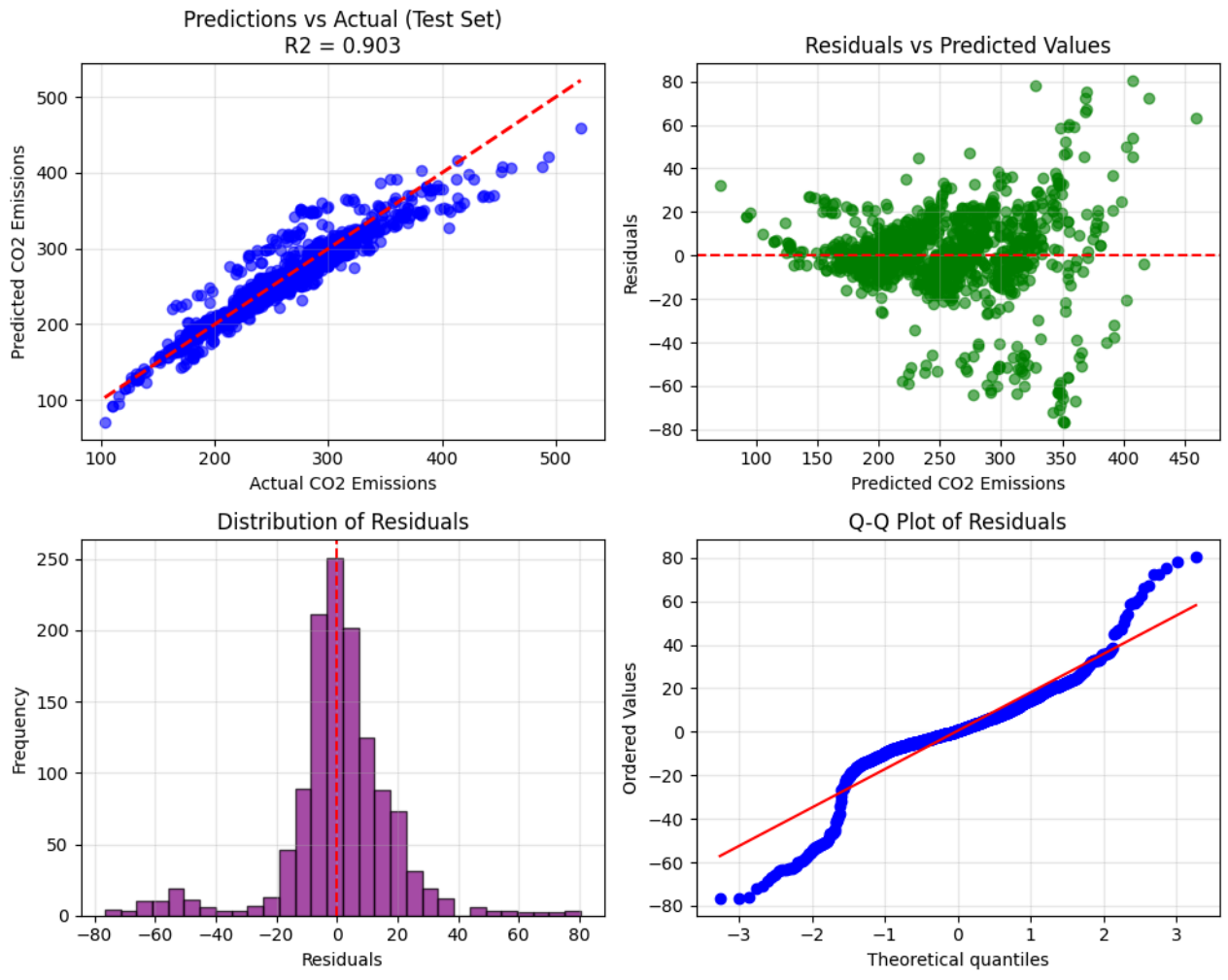
10. Model Evaluation and performance Assessment

Evaluate our models' performance using various metrics and visualization techniques.

```
# Make predictions
y_train_pred_full = lr_full.predict(X_train)
y_test_pred_full = lr_full.predict(X_test)

y_train_pred_selected = lr_selected.predict(X_train_selected)
y_test_pred_selected = lr_selected.predict(X_test_selected)

# Calculate performance metrics
def calculate_metrics(y_true, y_pred, dataset_name, model_name):
    r2 = r2_score(y_true, y_pred)
    mae = mean_absolute_error(y_true, y_pred)
    mse = mean_squared_error(y_true, y_pred)
    rmse = np.sqrt(mse)

    return {
        'Dataset': dataset_name,
        'Model': model_name,
        'R2': r2,
        'MAE': mae,
        'MSE': mse,
        'RMSE': rmse
    }

# Calculate metrics for all models and datasets
metrics_results = []

metrics_results.append(calculate_metrics(y_train, y_train_pred_full, 'Training', 'Full Features'))
metrics_results.append(calculate_metrics(y_test, y_test_pred_full, 'Testing', 'Full Features'))
metrics_results.append(calculate_metrics(y_train, y_train_pred_selected, 'Training', 'Selected Features'))
metrics_results.append(calculate_metrics(y_test, y_test_pred_selected, 'Testing', 'Selected Features'))

metrics_df = pd.DataFrame(metrics_results)

display(metrics_df.round(4))
```

	Dataset	Model	R2	MAE	MSE	RMSE
0	Training	Full Features	0.9174	11.1269	288.2847	16.9789
1	Testing	Full Features	0.9094	11.9241	326.9686	18.0823
2	Training	Selected Features	0.9074	11.5816	323.2523	17.9792
3	Testing	Selected Features	0.9027	12.1690	351.0050	18.7351

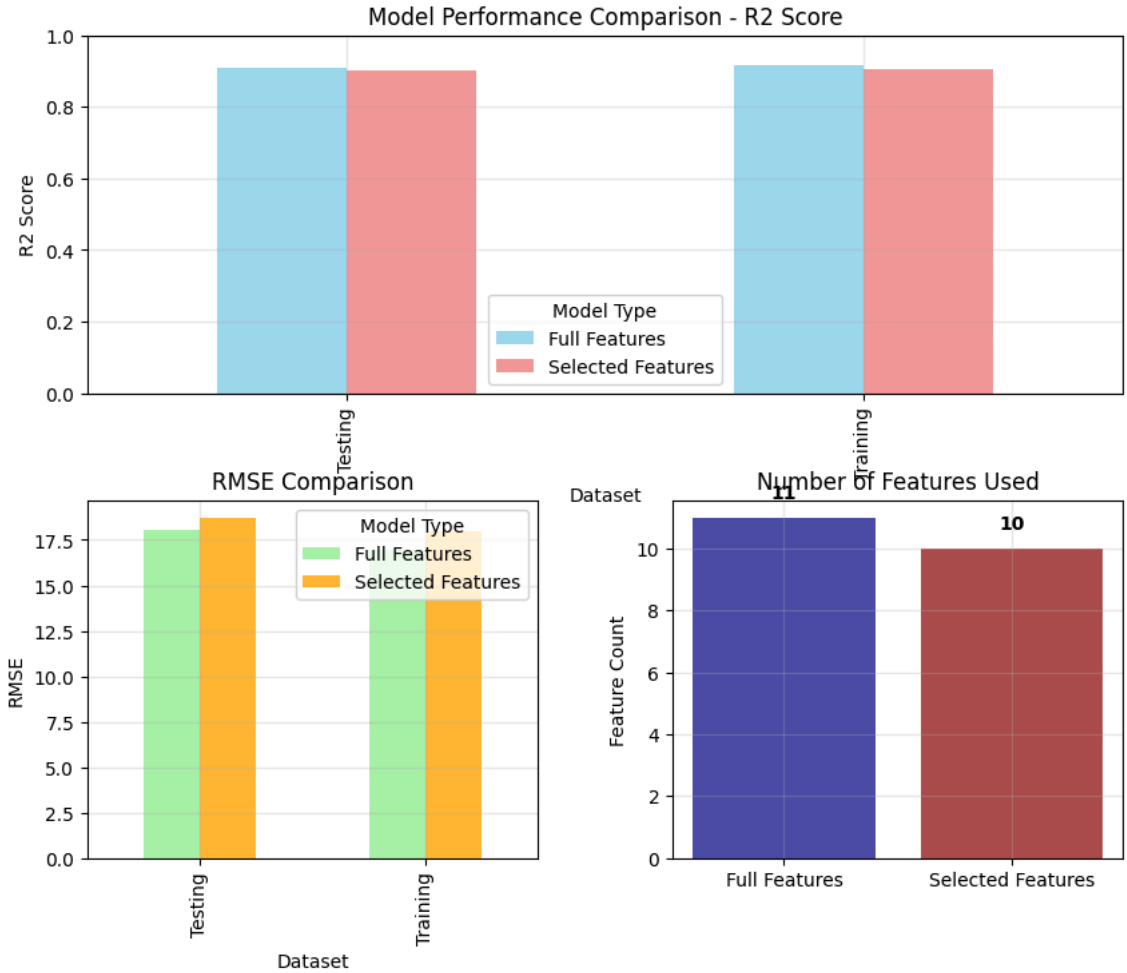
```
# Visualize model performance
fig, axes = plt.subplots(2, 2, figsize=(10, 8))

# Predictions vs Actual for Selected Features Model (Test Set)
axes[0,0].scatter(y_test, y_test_pred_selected, alpha=0.6, color='blue')
axes[0,0].plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()], 'r--', lw=2)
axes[0,0].set_xlabel('Actual CO2 Emissions')
axes[0,0].set_ylabel('Predicted CO2 Emissions')
axes[0,0].set_title(f'Predictions vs Actual (Test Set)\nR2 = {r2_score(y_test, y_test_pred_selected):.3f}')
axes[0,0].grid(True, alpha=0.3)

# Residuals plot
residuals = y_test - y_test_pred_selected
axes[0,1].scatter(y_test_pred_selected, residuals, alpha=0.6, color='green')
axes[0,1].axhline(y=0, color='red', linestyle='--')
axes[0,1].set_xlabel('Predicted CO2 Emissions')
axes[0,1].set_ylabel('Residuals')
axes[0,1].set_title('Residuals vs Predicted Values')
axes[0,1].grid(True, alpha=0.3)

# Distribution of residuals
axes[1,0].hist(residuals, bins=30, alpha=0.7, color='purple', edgecolor='black')
axes[1,0].set_xlabel('Residuals')
axes[1,0].set_ylabel('Frequency')
axes[1,0].set_title('Distribution of Residuals')
axes[1,0].axvline(x=0, color='red', linestyle='--')
axes[1,0].grid(True, alpha=0.3)

# Q-Q plot for residuals
stats.probplot(residuals, dist="norm", plot=axes[1,1])
axes[1,1].set_title('Q-Q Plot of Residuals')
axes[1,1].grid(True, alpha=0.3)

plt.tight_layout()
plt.show()
```




```
# Compare model performance visually
plt.figure(figsize=(10, 8))

# Create subplots for comparison
gs = plt.GridSpec(2, 2, hspace=0.3, wspace=0.3)

# R2 comparison
ax1 = plt.subplot(gs[0, :])
r2_data = metrics_df.pivot(index='Dataset', columns='Model', values='R2')
r2_data.plot(kind='bar', ax=ax1, color=['skyblue', 'lightcoral'], alpha=0.8)
ax1.set_title('Model Performance Comparison - R2 Score')
ax1.set_ylabel('R2 Score')
ax1.set_ylim(0, 1)
ax1.legend(title='Model Type')
ax1.grid(True, alpha=0.3)

# RMSE comparison
ax2 = plt.subplot(gs[1, 0])
rmse_data = metrics_df.pivot(index='Dataset', columns='Model', values='RMSE')
rmse_data.plot(kind='bar', ax=ax2, color=['lightgreen', 'orange'], alpha=0.8)
ax2.set_title('RMSE Comparison')
ax2.set_ylabel('RMSE')
ax2.legend(title='Model Type')
ax2.grid(True, alpha=0.3)

# Feature count comparison
ax3 = plt.subplot(gs[1, 1])
feature_counts = [X_train.shape[1], len(selected_features)]
model_names = ['Full Features', 'Selected Features']
ax3.bar(model_names, feature_counts, color=['navy', 'darkred'], alpha=0.7)
ax3.set_title('Number of Features Used')
ax3.set_ylabel('Feature Count')
ax3.grid(True, alpha=0.3)

# Add value labels on bars
for i, v in enumerate(feature_counts):
    ax3.text(i, v + 0.5, str(v), ha='center', va='bottom', fontweight='bold')

plt.tight_layout()
```

```
plt.show()
```

```
/tmp/ipykernel_172/2126142366.py:39: UserWarning: This figure includes Axes that are not compatible with tight_la
  plt.tight_layout()
```



Model Selection Recommendation:

Full Features Model - Test R2: 0.9094 Selected Features Model - Test R2: 0.9027 Performance difference: 0.0067

Recommended: Selected Features Model Reason: Similar performance with fewer features (better interpretability)

11. Business Insights and Recommendations

DATASET OVERVIEW:

• Total vehicles analyzed: 6,282 • Features examined: 12 • CO2 emissions range: 96.0 - 522.0 • Average CO2 emissions: 251.2

STRONGEST PREDICTORS OF CO2 EMISSIONS:

• Fuel Consumption City (L/100 km): Importance Score 27416.1 • Fuel Consumption Comb (L/100 km): Importance Score 26470.7 • Fuel Consumption Comb (mpg): Importance Score 23799.3 • Fuel Consumption Hwy (L/100 km): Importance Score 17690.2 • Engine Size(L): Importance Score 13606.6

Recommendations:

FOR AUTOMOTIVE MANUFACTURERS:

==========

1. REDUCE HIGH-IMPACT FACTORS: • Optimize Cylinders - each unit increase adds 7.26 emission units • Optimize Engine Size(L) - each unit increase adds 5.67 emission units • Optimize Fuel Consumption Hwy (L/100 km) - each unit increase adds 3.89 emission units
2. LEVERAGE EMISSION-REDUCING FACTORS: • Enhance Fuel Consumption Comb (mpg) - each unit increase reduces 3.34 emission units • Enhance Fuel Consumption Comb (L/100 km) - each unit increase reduces 1.84 emission units • Enhance Model - each unit increase reduces 0.00 emission units

3. DESIGN OPTIMIZATION PRIORITIES: • Focus R&D on the top 10 identified key factors • Target vehicles with characteristics similar to low-emission outliers • Benchmark against best-performing categories identified in analysis

FOR POLICYMAKERS:

==========

1. EMISSION STANDARDS: • Set progressive targets based on vehicle categories • Current average: 251.2, target 25th percentile: 208.0
2. INCENTIVE PROGRAMS: • Provide incentives for vehicles in lowest-emission categories • Penalize vehicles exceeding 289.0 emission threshold
3. MONITORING FRAMEWORK: • Use this model for emission prediction and verification • Focus inspections on vehicles predicted to have high emissions

FOR CONSUMERS:

==========

1. VEHICLE SELECTION GUIDANCE: • 2,102 vehicles (33.5%) in low-emission category • Choose vehicles with characteristics of identified low emitters • Avoid vehicle types identified as high emitters in categorical analysis